

Problem Statement 3: Coffee Shop Sales

Problem Statement

A multi-location coffee shop chain records every transaction — date, time, store, product, quantity, and price — but managers have no consolidated view of how sales perform across stores, products, and time of day. This makes it hard to spot which locations or product categories are underperforming, or when staffing and inventory should peak, without manually combing through raw transaction exports.

Objective: Build a Power BI report that gives store and operations managers a self-service view of sales performance by location, product, and time, so they can plan staffing and inventory more precisely.

Key business questions

1. Which store location and product category generate the most revenue?
2. What time of day and day of week see peak transactions, and does this vary by store?

Scope: Transaction-level sales data (transaction date/time, store location, product category/type, unit price, quantity); one Power BI report with an executive summary page, built using Power Query transformations and implicit (drag-and-drop) measures.

Data Acquisition and Preparation

Import the provided CSV into Power BI Desktop (Home → Get Data → Text/CSV → Transform Data) and perform the following Power Query transformations:

Step	Action
Change data type	Convert transaction_date to Date and transaction_time to Time
Remove duplicates	Check for and remove duplicate rows on transaction_id
Clean and trim	Trim/Clean extra spaces from store_location, product_category, and product_type
Replace null/blank values	Check product_detail for blanks; replace with “Cappuccino”
Add new columns	1. Add a Sales column = transaction_qty × unit_price

	2. Add new columns from the transaction date into day, month, year and quarter 3. Add new columns from the transaction_time into hour, minute
--	--

End with Close & Apply to load the cleaned data into the model.

Interactive Report and Executive Summary

All figures use implicit measures — drag a field into a visual and set the aggregation in the Values well. The Sales column added during data preparation feeds the KPI cards and charts below.

Executive Summary KPI cards

- Total Sales (Sum of Sales)
- Total Transactions (Count of transaction_id)
- Average Order Value (Average of Sales)
- Total Quantity Sold (Sum of transaction_qty)

Exploratory Data Analysis

- Sales by Store Location
- Sales by Product Category
- Transactions by Day of Week (derived from transaction_date)
- Revenue by hour of the day

Table graph

- Product Type, Category, Quantity Sold, Sales, Average Unit Price — sortable and filterable

Tooltip

- Build a tooltip page showing Product Detail and Average Unit Price when hovering over a product or category bar

Insights

1. Which store location and product category generate the most revenue?
2. What time of day / day of week sees peak transactions, and does it vary by store?

End with 1–2 concrete recommendations for store operations or staffing based on these findings.